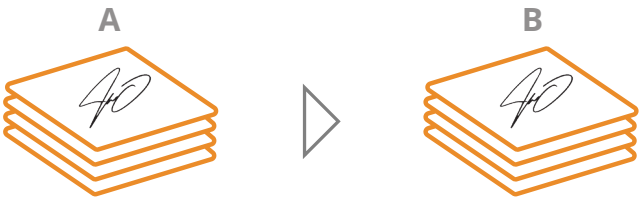
- Marc and Roby want to exchange 1 BTC. To understand this, it's important to know that there are no physical coins in ***bitcoin***, only updates to the *blockchain*, which then get reflected in the wallets of both parties involved.
- When Marc wants to send 1 BTC to Roby, it's called a **peer-to-peer** transaction because the ownership of the value goes directly from Marc to Roby. But Roby doesn't actually receive a "digital coin" from Marc. Instead, all the nodes in the network update their local copy of the public ledger, which changes the ownership of the ***bitcoin*** from Marc's address to Roby's address.
- A ***bitcoin*** **transaction** is simply a **signed message** that Marc sends to the network, which is then validated by many nodes. The message goes through several steps, like being picked up by some of the full nodes, being validated, and then being broadcasted, until all the nodes in the network have independently validated it.

The signature is a digital representation of the **transaction details**, including the amount of ***bitcoin*** being sent, the **sender's address (Marc)**, and the **recipient's address (Roby)**.

Purpose of a Digital Signature



A signature confirms that the message (document or email) originated from the sender and has NOT been modified.

Imagine all existing ***bitcoin*** as being stored in digital safes, each with a different amount of BTC along with a history of how they got there.

Each safe has an owner. Therefore, it has to be identifiable with an **address**. Addresses are protected by a digital lock with two different keys, like passwords to an account. If a safe has ***bitcoin***, its owner can open it with his **private key** and transfer any desired amount of ***bitcoin*** to another safe.

From Roby's perspective: To **receive** ***bitcoin***, you need to provide the sender (Marc) with your **address** where ***bitcoin*** can be deposited.

From Marc's perspective: To **spend** your ***bitcoin***, you need to access **your private key** for it to be unlocked.